

LEVEL 26 STRUCTURAL VERIFICATION — TATE, MAZUR, RIBET, KOLYVAGIN HONEST DISPLAYS IN LEAN 4.12

DAVID FOX

ABSTRACT. The Beal equation $A^x + B^y = C^z$ with $x, y, z > 2$ is studied here only for the restricted one-parameter family $C = B + 3$ inside the signature $(4, 4, 13)$, i.e. $A^4 + B^4 = (B + 3)^{13}$. This “gap-3” family is an internal naming convention of a larger, still largely open Lean development; it is *not* derived in this repository from the unrestricted equation $A^4 + B^4 = C^{13}$, and no claim about the general signature is made. For the test parameters $(A, B) = (29, 1)$ in the displayed Frey model — not a solution of the gap-3 equation itself — the associated Frey curve $E : y^2 = x(x - A^4)(x + B^4)$ has displayed conductor $928 = 2^5 \cdot 29$, Néron types I_0^* at 2 and I_8 at 29, and valuations $v_2(\Delta) = 6$, $v_{29}(\Delta) = 8$; Tate’s algorithm itself is not formalized in Mathlib 4.12, and the same family at $(A, B) = (1, 1)$ displays conductor 32, not 928. This draft records the Lean 4.12 verification of four families of displayed numeral and Finset facts, named only by association with Tate, Mazur, Ribet, and Kolyvagin; none of the four classical theorems is proved. Mazur-named display: $|\mathrm{SL}_2(\mathbb{F}_{13})| = 2184$, $48 < 2184$, $13 = 2^2 + 3^2$, $288/48 = 6$, cusp Finset card 2; $X_0(13)$ has genus 0, so $X_0(13)(\mathbb{Q})$ is infinite and no 13-isogeny is excluded by this display. Ribet-named display: $928/29 = 32$, an empty matching Finset (card 0) against the LMFDB dimension-1 newform space at level 32 (32a1). Kolyvagin-named display: $|\mathrm{Sel}_2| = 1$, $3 \cdot 7 = 21$, $L/\Omega = 1/3$ and $1/7$ on Cremona 26a1/26b1, and the typeclass upgrade $\mathrm{Nonempty}(\mathrm{Fintype} \alpha) \rightarrow \mathrm{Finite} \alpha$; the repository’s surrogate `IsSubsingleton := Subsingleton` (renamed from `IsRankZero` in this revision) is not the conventional algebraic-rank-zero predicate, and its application to each curve is false, since both have nontrivial torsion. Separately, the repository proves a genuine finite lemma, `no_match_32a1_of_good_red_29`: on the gap-3 equation with $29 \nmid A$, $29 \nmid B$, $29 \nmid B + 3$, the Frey trace at 29 cannot equal -10 , the trace of 32a1, by an exhaustive residue check compiled with `native_decide` (axiom `Lean.ofReduceBool`). This lemma is not connected in Lean to Ribet’s level-lowering step, which stays an uninhabited `def Prop`, and it does not address $29 \mid ABC$. The four named displays themselves use explicit types and `decide`, not `native_decide`. This revision’s version DOI, 10.5281/zenodo.22885060, was reserved through Zenodo’s deposit API before publication so that this sentence could cite it correctly (Section 7 explains why three earlier v31 records could not); it supersedes 10.5281/zenodo.22863527 for v30.0.0-level-26-structural-verification, which still archives the pre-rename declaration names discussed above. Concept DOI 10.5281/zenodo.22041831 (this repository’s own concept series; the separate `beal-level-26-foundations` repository has its own concept DOI, 10.5281/zenodo.22379293, with final forward-interface version 10.5281/zenodo.22912430 and post-final v25 retrofit 10.5281/zenodo.22922473). The retrofit orders the Matveev bound, level-26 arithmetic, proved $B < 1000$ slice, and explicit $B \leq 10^6$ boundary without using numerical equalities backwards.

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1. INTRODUCTION

The Beal conjecture asserts that if $A^x + B^y = C^z$ with integers $A, B, C \neq 0$ and exponents $x, y, z > 2$, then A, B, C share a prime factor [11]. This draft does not prove the conjecture, does not address the unrestricted signature $(4, 4, 13)$ equation

$$A^4 + B^4 = C^{13},$$

and does not derive any reduction of that equation. It records a *structural verification* of Lean 4.12 material attached to a single, deliberately restricted one-parameter family inside that signature, together with an honest account of what each Lean name does and does not establish.

The gap-3 family: what it is and where it comes from. Throughout, “the gap-3 equation” means

$$A^4 + B^4 = (B + 3)^{13}, \tag{1}$$

i.e. $C := B + 3$. This is *not* a reduction proved from the unrestricted equation anywhere in the repositories accompanying this draft. It is the name given, inside a much larger and mostly uninhabited Lean development (files `BealGap3BakerUpperBound.lean`, `BealFreyB14.lean`, `DarmonMerelFrey4413.lean`, and the `Beal_4_13_13_Zsigmondy_13_Step*` series), to one specific test family that the author chose to formalize as a vehicle for exercising Tate/Mazur/Ribet/Kolyvagin-style arguments in Lean. Four points, requested explicitly by an external reviewer, are recorded here rather than left implicit:

- (1) *Where $C = B + 3$ comes from.* It is an adopted normalization, not a derived one. The surrounding Lean files contain a size lemma for the *different* triple $(4, 13, 13)$ showing $C^{13} - B^{13} \geq 26B^{12}$ when $C \geq B + 2$ (`Beal_4_13_13_Size_C_ge_B_plus_2.lean`) and an observation that $\gcd(B, B + 3) \mid 3$ (`gcd_B_Bplus3_dvd_three` in `BealFreyB14.lean`), but neither of these, nor anything else found in the repository, is a theorem that a hypothetical $(4, 4, 13)$ counterexample must have $C - B = 3$. The gap-3 family is a case study, not a WLOG reduction.
- (2) *What it covers.* Only the one-parameter slice (1), for arbitrary $A, B \in \mathbb{N}$. It says nothing about $C - B \in \{1, 2, 4, 5, \dots\}$ or about solutions, if any, of the unrestricted equation.
- (3) *Whether the “good reduction at 29” hypotheses hold in any intended case.* They do not hold automatically; they are hypotheses of the lemma discussed in Section 4, and no Lean theorem in the repository shows that a hypothetical solution of (1) must satisfy $29 \nmid ABC$. The case $29 \mid ABC$ is untouched.
- (4) *Whether a trace mismatch eliminates the level-32 newform after level lowering.* It does not, in Lean, because the level-lowering step itself (`ribet_lowers_Frey_to_level_32`) stays an

uninhabited **def Prop**: there is no formal link between the actual Galois representation of the Frey curve and the newform 32a1 whose trace is being compared.

Section 4 returns to this with the specific lemma statement, its hypotheses, and its axiom profile.

The strategy, honestly labelled. The classical strategy this development is modeled on is Frey–Serre–Ribet–Mazur [8, 14, 13, 12]: compute the conductor of a Frey curve by Tate’s algorithm [15], exclude a rational 13-isogeny by Mazur’s theorem on $X_0(13)$, lower the residual level in the style of Ribet, and compare the resulting newform space with Mordell–Weil data via the Cremona labels 26a1 and 26b1 [1, 10]. None of those four classical theorems is proved, used, or even stated as a hypothesis-free implication anywhere in the Lean files discussed below. The Frey curve of (1) is the displayed Weierstrass model

$$y^2 = x(x - A^4)(x + B^4) \quad \text{i.e.} \quad [0, B^4 - A^4, 0, -A^4 B^4, 0]. \quad (2)$$

The short model $[0, -(A^4 + B^4), 0, 0, 0]$ is a different curve and is not used as E . Sections 2–5 are titled “displayed data” for exactly this reason: each records what four Lean files, named *after* the classical authors, actually contain — numerals, Finset cardinalities, and one typeclass upgrade — while the corresponding algebraic statements of Tate/Mazur/Ribet/Kolyvagin remain uninhabited **def Prop** in the parent files.

Pins and provenance. The formal pin is Lean v4.12.0, mathlib commit 809c3fb3b5c8f5d7dace56e200b426187516535a with inputRev v4.12.0 (not @ main), and the beal-conjecture Lake manifest prefix 83739542. The command lake update was not run. The vendor tree Level26/BealLevel26Foundations at db7a556/fea0c393 has difference 0 against b780c9c/1a00fbd. This revision’s version DOI, 10.5281/zenodo.22885060, was reserved through Zenodo’s deposit API before publication (Section 7); it is a new version of 22863527 (v30), which was itself a new version of 22854603 and 22851334; the beal-conjecture concept DOI 10.5281/zenodo.22041831 stays, related as isVersionOf [4, 2]. The separate beal-level-26-foundations repository has its own concept DOI, 10.5281/zenodo.22379293 [3]; it is not the same series as this repository’s. Its end-of-proof release v30.1.7-beal-final-43735b3 is archived as 10.5281/zenodo.22912430, and the post-final v25.0.1-forward-43735b3 retrofit as 10.5281/zenodo.22922473 [6, 7].

This is a Mathematics of Computation manuscript draft, not an arXiv preprint. There is no arXiv identifier and no arXiv badge.

The four display files live in Level26/HonestB0Search/ on beal-conjecture main (release tag v30.0.0-level-26-structural-verification, commit af520d3) and on foundations phase-darmon-merel-4413 at 2b60ccd. At this writing the HonestB0Search slices are near identical; beal-conjecture is the superset, carrying the historical BealMatveevBeal target in addition to the four finals. Any commit hash for main cited elsewhere in this draft for purposes other than pinning a specific historical file is only accurate as of the revision that introduced it; the release tag above is the stable anchor.

2. DISPLAYED DATA I — A TATE-STYLE CONDUCTOR NUMERAL AT $(A, B) = (29, 1)$

The file Tate_Frey_Conductor_29_Neron_final.lean (11c0ad7; foundations incremental 35488239286, 137s cache hit) packages the displayed conductor numeral, for the test parameters $(A, B) = (29, 1)$ in the displayed Frey model (2) associated with the gap-3 development. These are test parameters for the model, not a solution of the gap-3 equation (1): $29^4 + 1^4 \neq (1 + 3)^{13}$.

$$N_E^{\text{disp}} = (2 : \mathbb{N})^5 \cdot 29 = 928.$$

The named theorems displayed_Neron_conductor and final_N_E_eq_928 are that equality, proved by decide. The split displayed_928_div_29 is $928/29 = 32$. At the other named specialization $(A, B) = (1, 1)$ the same family displays conductor 32, not 928

(`sagemath/certs/tate_nero_29.json`): the numeral 928 is a fact about one pair (A, B) , not a property of every solution of (1).

Display 2.1 (Displayed Néron conductor). In Lean 4.12, with `open Nat Finset Classical`:

- $(2 : \mathbb{N})^5 \cdot 29 = 928$ (`final_N_E_eq_928`);
- $928/29 = 32$ (`displayed_928_div_29`);
- $32 \cdot 29 = 928$;
- $\neg 29 \mid 32$.

These are numerals. Mathlib 4.12 has no `KodairaType` and no `NeronModel`. The name $N_E = 928$ is not the output of a Lean formalization of Tate’s algorithm or of a Néron model.

The displayed Weierstrass invariants of (2) are

$$c_4 = 16(A^8 + B^8 + A^4B^4), \quad \Delta = 16A^8B^8(A^4 + B^4)^2.$$

On the inhabited slice one has $v_2(\Delta(1, 1)) = 6$ (type I_0^* valuation) and $v_{29}(\Delta(29, 1)) = 8$ (type I_8), together with $29 \nmid c_4(29, 1)$. The residual-level Finset $\{32, 928\}$ has card 2; that card is a Finset numeral over two named specializations, not two Néron models of a single curve. The single named theorem `displayed_frey_specialization_data_at_2_and_29` re-exports this inhabited conjunction, i.e. `Tate_algorithm_at_2_and_29`.

Honesty note 2.2 (v31 rename). Earlier releases (through v0.30) additionally exported this same conjunction under three further names, `Tate_algorithm_at_29`, `Frey_Neron_conductor`, and `Frey_conductor_29_is_Neron`, chosen to echo Tate’s algorithm and Néron conductors even though the file computes neither. Those names, and a fourth packaged theorem of the same type once named `Tate_Frey_Conductor_29_Neron_final`, are removed as of this revision; there is no compatibility alias. The differently-scoped parent names `Tate_algorithm_at_29`, `Frey_Neron_conductor`, and `Frey_conductor_29_is_Neron` on `Tate_Frey_Conductor_29.lean` are untouched and remain `def Prop`.

Honesty note 2.3. The phrase “final $N_E = 928$ ” means the displayed identity $2^5 \cdot 29 = 928$ at $(A, B) = (29, 1)$. It is not Tate’s algorithm as a function on integral models, and it is not a claim about the conductor for other (A, B) satisfying (1). SAGE `sagemath/tate_nero_29.sage` uses the displayed Frey (2), not the short model, and its own comment records that $(1, 1)$ gives 32.

Axioms in this file are `[propext]` for the pure numerals (e.g. `final_N_E_eq_928`) and `[propext, Classical.choice, Quot.sound]` for `displayed_frey_specialization_data_at_2_and_29`, which mentions valuations already inhabited upstream. There is no `Lean.ofReduceBool`, no `sorryAx`, and no new axiom.

3. DISPLAYED DATA II — MAZUR-STYLE NUMERALS FOR $X_0(13)$

The file `Mazur_X0_13_No_Isogeny_final.lean` (2f4e16e; incremental 35488699511, 252s) displays numerals associated with $X_0(13)$ [12]. None of them excludes a rational 13-isogeny; see the honesty note below.

Display 3.1 (Mazur numerals). • $|\mathrm{SL}_2(\mathbb{F}_{13})| = 13 \cdot 12 \cdot 14 = 2184$ (`SL2_F13_card`);

- $48 < 2184$ (`forty_eight_lt_2184`);
- $13 = 2^2 + 3^2$ (`thirteen_eq_two_squares`);
- $288/48 = 6$ (`two_eighty_eight_div_forty_eight`);
- $(\{0, 1\} : \text{Finset } \mathbb{N}).\text{card} = 2$ (`displayed_cusps_card`).

The first four identities are axiom-free `decide` theorems. The cusp-card theorem uses `[propext, Quot.sound]`.

The single named theorem `displayed_X0_13_numerical_data` re-exports the inhabited conjunction of Display 3.1 with the genus Nat 0; despite the name, it does not state or prove that a rational 13-isogeny is excluded.

Honesty note 3.2 (Genus 0 versus two cusps, and no isogeny exclusion). $X_0(13)$ has genus 0, so $X_0(13)(\mathbb{Q})$ is infinite over $\mathbb{Q}$. The pasted equality $\{2 \text{ cusps}\} = X_0(13)(\mathbb{Q})$ is literature-false and is *not* a theorem. The honest Lean display is finite card 2 versus the genus-0 note `X0_13_genus_nat ≠ X0_13_cusp_count`, packaged in `displayed_X0_13_numerical_data`; it is not a Mathlib proof that the rational points are infinite, and it is not Mazur’s isogeny theorem. In particular, the introduction’s classical strategy paragraph names Mazur’s theorem for context only: no Lean theorem in this repository excludes a rational 13-isogeny of the Frey curve, and the genus-0/infinity fact recorded here is logically unrelated to such an exclusion.

v31 rename. Earlier releases (through v0.30) exported this same conjunction four further times, under the names `X0_13_Q_infinite`, `frey_no_rational_13_isogeny`, `Serre_non_Borel_mod13`, and `mazur_no_Frey_13_isogeny`, plus a fifth, slightly smaller packaged theorem once named `Mazur_X0_13_No_Isogeny_final`. Those names invited a reader to mistake the numeral display for an isogeny exclusion, a non-Borel image statement, or Mazur’s theorem itself. They are removed as of this revision; there is no compatibility alias. The differently-scoped parent names on `Mazur_X0_13_No_Isogeny.lean` are untouched and remain `def Prop`; it is that parent `Prop`, not any theorem in the final-display file, that would correspond to an actual exclusion statement.

Geometrically, a genus-0 curve has trivial Jacobian, so $J_0(13)$ is a point (rank 0, finite). That remark is not claimed as a Lean Mordell–Weil theorem in the display file. Mazur’s theorem that the only rational 13-isogenies of elliptic curves over $\mathbb{Q}$ arise from the cusps remains `def Prop` on the parent, uninhabited.

Axioms: numerals `[propext]` or axiom-free; package `[propext, Quot.sound]`; LLL no-go persistence `[propext, Classical.choice, Quot.sound]`. No `native_decide`, no `sorryAx`, no new axiom.

4. DISPLAYED DATA III — RIBET-STYLE NUMERALS FOR LEVEL 32, AND A CONDITIONAL TRACE LEMMA AT 29

The packaged display. The file `Ribet_Level_Lowering_29_to_32_final.lean` (5631fd1; incremental 35489274297, 282s) displays the residual-level arithmetic named after Ribet [13] without inhabiting abstract level lowering.

Display 4.1 (Ribet numerals). • $928/29 = 32$, $32 \cdot 29 = 928$, $\neg 29 \mid 32$;

- $2^4 = 16$, $16 \mid 32$;
- $\text{displayed dim } S_2(\Gamma_0(32)) = 1$, $\text{displayed dim } S_2(\Gamma_0(16)) = 0$;
- Sturm pair $32 \cdot \frac{3}{2} = 48$ and $2 \cdot 48/12 = 8$;
- $(\emptyset : \text{Finset } \mathbb{N}).\text{card} = 0$.

The displayed full/new dimensions and the Sturm pair are the LMFDB pin `full=1`, `new=1`, `level16=0`, `sturm=8` for the unique newform 32a1 [10].

The single named theorem `displayed_level32_arithmetic_and_newspace_data` re-exports `no_newforms_at_32`: the numerals of Display 4.1 together with `displayed_frey_matching_newforms.card = 0`. That `Finset` is defined by `:= ∅`, not computed from the Frey curve’s traces, so its cardinality theorem is immediate and proves nothing about which newform the Frey curve matches.

Honesty note 4.2 (v31 rename). Earlier releases (through v0.30) exported this same conjunction under three further names, `Ribet_928_to_32`, `no_newforms_at_32_mod13`, and `explicit_a29_mod13`, plus a fourth packaged theorem of the same type once named `Ribet_Level_Lowering_29_to_32_final`. The name `no_newforms_at_32_mod13` was especially misleading,

since LMFDB records a one-dimensional newspace at level 32, and `explicit_a29_mod13` did not compute a_{29} . All four are removed as of this revision; there is no compatibility alias. The differently-scoped parent names on `Ribet_Level_Lowering_29_to_32.lean` are untouched and remain `def Prop`. The genuinely computed lemmas discussed below, `no_match_32a1_of_good_red_29` and `no_sol_with_good_red_29_of_trace_match`, describe their actual conditional conclusions and are kept unchanged.

Honesty note 4.3 (Newspace dimension 1, not 0). LMFDB records $\dim S_2(\Gamma_0(32))^{\text{new}} = 1$, with unique newform 32a1. The pasted Sage line `CuspForms(32,2).new_subspace().dimension()` is *not* a theorem. The honest Lean display is the empty *asserted* matching Finset (card 0) versus displayed dimension 1; it is a mismatch of definitions, not a derived elimination. Witnesses: `sagemath/ribet_no_newforms_32.sage` and `certs/ribet_no_newforms_32.json` pin `full=1`, `new=1`, `level16=0`, `sturm=8` (`sage_ran: false` in the JSON: these are recorded, not freshly computed, values).

Axioms on the packaged display: `[propext, Quot.sound]`. LLL no-go persistence: `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom for either.

A separate, genuinely computed lemma: `no_match_32a1_of_good_red_29`. The file `Level32Table.lean`, imported by `Ribet_Level_Lowering_29_to_32_final.lean` but not used by any of the theorems above, contains a lemma that *is* derived rather than asserted:

$$\begin{aligned} & \text{no_match_32a1_of_good_red_29 } \{A\ B : \mathbb{N}\} \\ & (\text{hsol} : A^4 + B^4 = (B + 3)^{13}) (\text{hA} : 29 \nmid A) (\text{hB} : 29 \nmid B) (\text{hC} : 29 \nmid B + 3) : \\ & \text{frey_ap } 29 (A \bmod 29) (B \bmod 29) \neq (-10 : \mathbb{Z}). \end{aligned} \quad (3)$$

Here `frey_ap p A B` is the trace $a_p = p - \#E_{A,B}(\mathbb{F}_p)$ of the naive reduction of (2) mod p , computed by counting affine points, and `-10` is `curve32a1_ap_29`, the a_{29} coefficient of 32a1. The proof reduces (3) to a closed instance, `frey_a29_ne_32a1_fin`: for every pair $(A, B) \in (\mathbb{Z}/29\mathbb{Z})^2$ with $A \neq 0$, $B \neq 0$, $A^4 + B^4 \not\equiv 0 \pmod{29}$, and $A^4 + B^4 \equiv (B + 3)^{13} \pmod{29}$ — i.e. every residue pair consistent with a good-reduction solution of (1) at 29 — the trace is $\neq -10$. This is a finite check over the $29^2 = 841$ candidate pairs, proved by `native_decide`, not `decide`.

Honesty note 4.4 (Axiom profile differs from the four named displays). Rebuilding `Level32Table.lean` on the pinned toolchain and running `#print axioms no_match_32a1_of_good_red_29` reports

`[propext, Classical.choice, Lean.ofReduceBool, Quot.sound]`.

The presence of `Lean.ofReduceBool` is exactly the mark of `native_decide`: this lemma relies on Lean’s trusted native-evaluation bridge `Lean.ofReduceBool`, rather than on reduction of the full finite check over $29^2 = 841$ residue pairs by the kernel, unlike every named-display theorem in Sections 2–5, which are stated above to avoid `native_decide` and to have no `Lean.ofReduceBool` in their axiom lists. Both facts are true simultaneously: the four packaged `_final` displays really do avoid `native_decide`, and this separate, uncited lemma really does use it. Citing `no_match_32a1_of_good_red_29` as part of an axiom-clean argument would be inaccurate.

Under its stated hypotheses — a gap-3 solution with good reduction at 29 on all of A , B , and $B + 3$ — the lemma is a real, mechanically checked mod-29 trace mismatch, and the assessment in this draft’s first version understated it. It does not, however, complete an elimination argument for three reasons recorded in Section 1: (i) it says nothing about $29 \mid ABC$; (ii) no Lean theorem shows a hypothetical solution of the gap-3 family must have good reduction at 29, so the hypothesis is not known to hold in any case of interest; and (iii) the identification of a_{29} of the Frey curve’s

Galois representation with a_{29} of 32a1 — which is what a completed Ribet/level-lowering argument would supply — is not derived here; `ribet_lowers_Frey_to_level_32` stays an uninhabited `def Prop` (Section 6). The companion theorem `no_sol_with_good_red_29_of_trace_match` makes this last gap explicit: it takes the trace-match hypothesis that Ribet would need to supply as an additional argument and derives `False` from it via `no_match_32a1_of_good_red_29`. Its axiom list, `[propext, Classical.choice, Lean.ofReduceBool, Quot.sound]` (checked the same way), still carries `Lean.ofReduceBool`, since it calls the `native_decide`-based lemma directly rather than merely citing its statement; no theorem in this file removes that dependency.

5. DISPLAYED DATA IV — KOLYVAGIN-STYLE NUMERALS ON 26A1, 26B1

The file `Kolyvagin_MW_Rank0_26a1_26b1_final.lean` (fc8c7a1 / phase merge bc7e93b; incremental 35489926310, 262s) displays numerals for Cremona 26a1 (LMFDB 26.a2) and 26b1 (LMFDB 26.b2) [9, 1, 10].

Display 5.1 (Kolyvagin numerals and typeclass upgrade). • $|\text{Sel}_2|^{\text{disp}} = 1$ (`Sel2_card_one`, `displayed_Sel2_card_26a1/b1`);

- $3 \cdot 7 = 21$ (`three_mul_seven`);
- $L/\Omega = 1/3$ on 26a1, $1/7$ on 26b1 (`norm_num` on $\mathbb{Q}$);
- $\neg \text{IsSubsingleton}$ on both models;
- $\text{Nonempty}(\text{Fintype } \alpha) \rightarrow \text{Finite } \alpha$ (`finite_iff_nonempty_fintype`);
- $|\text{Unit}| = 1$ (`Subsingleton card on Unit`).

The rationals $3/9 = 1/3$ and $7/49 = 1/7$ use `norm_num`: `decide` has no $\mathbb{Q}$ instance in Lean 4.12.

Cremona 26a1 has torsion $\mathbb{Z}/3\mathbb{Z}$ and $\Delta = -17576$; 26b1 has torsion $\mathbb{Z}/7\mathbb{Z}$ and $\Delta = -1664$. The repository’s surrogate `IsSubsingleton := Subsingleton` (renamed, this revision, from `IsRankZero`; see below) is not the conventional algebraic-rank-zero predicate, and its application to each of these curves is false, because each curve has nontrivial torsion (the theorems `not_IsSubsingleton_26a1_final` and `not_Subsingleton_MW_26a1_final`, and the 26b1 twins). The `Subsingleton → Fintype` path applies to `Unit`, not to $E(\mathbb{Q})$. The section title above says “numerals,” not “rank 0,” precisely because the repository’s own display file proves `IsSubsingleton` *fails* for the curves it names.

The single named theorem `displayed_26a1_26b1_certificate_data` packages Display 5.1: $\neg \text{IsSubsingleton}$ on both curves, $|\text{Sel}_2| = 1$, $3 \cdot 7 = 21$, L/Ω , and the typeclass upgrade.

Honesty note 5.2 (Not $E(\mathbb{Q})$ finite). Algebraic Mordell–Weil rank 0 as `Nonempty(Fintype(E(Q)))` remains `def Prop` on the parent (`Kolyvagin_MW_Rank0_26a1_26b1.lean`). The packaged theorem `displayed_26a1_26b1_certificate_data` is Display 5.1, not Kato–Kolyvagin and not Sage `E.rank()`. Witness: `sagemath/kolyvagin_rank0_26.sage` with `certs/kolyvagin_rank0_26.json` (`sage_ran: false`: the file records pinned values, not a fresh Sage computation).

Honesty note 5.3 (v31 rename). Earlier releases (through v0.30) exported this same conjunction seven further times: as the parent-named theorems `MW_rank_zero_fintype`, `Kolyvagin_L_nonzero_imp_MW_rank_zero`, and `TwoDescent_implies_MW_rank_zero_fintype` (duplicating uninhabited `def Prop` names on a different parent file), and as the module-named theorems `TwoDescent_26a1_26` (from 7c19ad0), `BSD_MordellWeil`, `J0_26_BSD_26a1_26b1`, and `X0_26_Full12Torsion` (duplicating the names of separate, genuinely different modules), plus an eighth packaged theorem of the same type once named `Kolyvagin_MW_Rank0_26a1_26b1_final`. All eight invited a reader to mistake this numeral display for the parent’s uninhabited claim, for BSD, or for the named module. They are removed as of this revision; there is no compatibility alias. On the parent files, `MW_rank_zero_fintype`, `Kato`, `BSD_MordellWeil.MW_rank_zero`, and `X0_26_Full12Torsion.J0_26_rank0` are untouched and remain `def Prop`. Separately,

the definition `IsRankZero := Subsingleton` on `BSD_MordellWeil.lean` is renamed to `IsSubsingleton`, since “rank zero” conventionally permits nontrivial finite torsion while `Subsingleton` does not; all theorems whose names referenced `IsRankZero` are renamed to match (e.g. `not_IsRankZero_26a1_final` is now `not_IsSubsingleton_26a1_final`).

Axioms: `[propext, Classical.choice, Quot.sound]`. No `Lean.ofReduceBool`, no `sorryAx`, no new axiom.

6. LEAN FORMALIZATION — EXPLICIT TYPES

Every theorem in the four finals carries an explicit type. Lean 4.12 rejects `theorem foo := bar`. The required opens are `open Nat Finset Classical`. Typical types are `ℕ`, `Prop`, `Finset ℕ`, and `Fintype`. Proofs use `decide` (or `norm_num` on `ℚ`), never `native_decide`, so `Lean.ofReduceBool` does not appear.

Pin. Toolchain `leanprover/lean4:v4.12.0`. Mathlib commit `809c3fb3b5c8f5d7dace56e200b426187516535a`, inputRev `v4.12.0`. Foundations Lake manifest prefix `c7dbd578`; conjecture prefix `83739542` (package lists differ; do not run `lake update`). Vendor `db7a556/fea0c393` has difference 0 against `b780c9c/1a00fbd` and against foundations `43735b3`.

Axioms. For the four packaged `_final` displays of Sections 2–5, allowed axiom lists are `[propext]`, `[propext, Quot.sound]`, and `[propext, Classical.choice, Quot.sound]`. There is no `sorryAx` and no new axiom, and `Lean.ofReduceBool` does not occur, i.e. none of the four uses `native_decide`. Parent files keep the algebraic names as `def Prop`. Display files contain no new `def Prop`. This does *not* extend to every theorem in every file those four displays import: Section 4 records one lemma, `no_match_32a1_of_good_red_29` in `Level32Table.lean`, whose axiom list is `[propext, Classical.choice, Lean.ofReduceBool, Quot.sound]` because its proof uses `native_decide` on a finite residue check. Both were verified on the pinned toolchain with `lake build +Level32Table` followed by the file’s own `#print axioms` lines (cache warm, ~ 1.3 s), and, for the companion theorem `no_sol_with_good_red_29_of_trace_match`, with a one-line standalone `lake env lean check (import Level32Table; #print axioms no_sol_with_good_red_29_of_trace_match)`, which also reports `Lean.ofReduceBool`.

Integrated end-of-proof and v25 forward interfaces. The companion foundations releases preserve a six-stage Tate–Mazur–Ribet–Kolyvagin–Conditional-32–Requires ledger in `BealFinal.Main`. The general Beal common-prime conclusion is carried as explicit proof data rather than inferred from the signature $(4, 4, 13)$ numerical surface.

The post-final module `BealMatveevBealV25B0Search.Forward` orders `MatveevBoundData`, `Level26SearchData`, the proved $B < 1000$ elimination, and the explicit full $B \leq 10^6$ proposition. Its numerical projection records $\Delta = 16A^8B^8(A^4 + B^4)^2$, $v_{29}(\Delta) = 26v_{29}(B + 3)$, $13 \mid v_{29}(\Delta)$, $928/29 = 32$, and $32 \cdot 29 = 928$. None of those identities constructs the Matveev, level-lowering, or full-search boundaries. The three retrofit projections report exactly `[propext, Classical.choice, Quot.sound]`.

CI, incremental only. Foundations `build.yml` is `HonestB0Search` incremental: `lake exe cache get`, then `+Tate_Frey_Conductor_29_Neron_final`, `+Mazur_X0_13_No_Isogeny_final`, `+Ribet_Level_Lowering_29_to_32_final`, `+Kolyvagin_MW_Rank0_26a1_26b1_final` (~ 40 s on a warm `.lake`), then `lake build HonestB0Search` and `lake build Level26` (1–2s each). Cache key `lake- $\{hashFiles(lean-toolchain, lake-manifest.json)\}$ -v27-6ccafbf`. Representative greens: `35488239286` (137s, `11c0ad7`); `35488699511` (252s); `35489274297` (282s); `35489926310` (262s); phase `35490246507` (303s) and `35524223408`; About chore `35552037733` (01:47:20Z–01:52:05Z, 285s, `.lake` restored). On those working branches `main.yml` recorded 0 runs. The phase incremental `35524223408` is the green gate. This draft does not run `lake build BealMatveevBeal` and does not run `lake update`.

Conjecture `main.yml`. On `beal-conjecture` the hours job `build+HonestB0Search+Level26` runs only on `main` (and matching `merge/phase/v24-v27` branches). Greens: `merge-branch 35524441913 (850e1e9)`; `main 35532850099 (af520d3)`; badge `35541392250 (22:20:55Z-00:58:37Z, 2h 37m)` on `ff546da`. Tip `ea261b9` is the About chore on top of that badge.

7. STATIC COMPUTATIONAL CERTIFICATES AND KERNEL CHECKS

Zenodo. The durable citation for this software is the `beal-conjecture` concept DOI, `10.5281/zenodo.22041831` [2], which always resolves to whichever version is most recently archived under it; cite it when the exact version does not matter. This revision’s own version DOI, `10.5281/zenodo.22885060`, was reserved through Zenodo’s deposit API *before* this paragraph was written and published only once the source tree — including this sentence — matched the reservation, so, unlike the two records below, it is not one commit behind its own citation. It supersedes `10.5281/zenodo.22884873` (tag `v31.0.1-level-26-honest-declarations`, commit `b5c8911`) and `10.5281/zenodo.22884682` (tag `v31.0.0-level-26-honest-declarations`, commit `0367d48`), both left live as historical intermediate versions, which in turn supersede `10.5281/zenodo.22863527` (`v30.0.0-level-26-structural-verification`, GitHub release id `392549396` at `af520d3`), itself a new version of `22854603` and `22851334` [4, 5]. The `22863527` snapshot archives the pre-rename Lean numerals discussed in Sections 2–5, not the renamed declarations. The separate `beal-level-26-foundations` repository has its own concept DOI, `10.5281/zenodo.22379293` [3], whose `v0.31` record is `10.5281/zenodo.22884683`. The later end-of-proof release `v30.1.7-beal-final-43735b3` is `10.5281/zenodo.22912430`; its post-final `v25` forward retrofit is `10.5281/zenodo.22922473`. These records belong to the foundations concept series, not to `beal-conjecture`’s `22041831` series.

Honesty note 7.1 (Three prior inaccuracies about this Zenodo archival, found and fixed while writing this revision). First, an earlier draft of this section cited `v31` identifiers that were Git tags with no corresponding GitHub Release object, so Zenodo’s GitHub integration — which archives *releases*, not bare tags — had not minted a version DOI for either repository; fixed by publishing `22884682`. Second, every earlier draft of this paper, going back to the first `v30` draft, stated that `beal-conjecture` and `beal-level-26-foundations` share one concept DOI, `10.5281/zenodo.22379293`; checking the public Zenodo API showed that is false, so `22041831` is cited above instead. Third, the commit fixing the second inaccuracy (`b5c8911`) was itself published as `22884873` *before* that commit existed to be archived, since a GitHub-Release-triggered Zenodo mint necessarily happens strictly after the triggering commit is pushed: no automatically-triggered archive can ever cite its own DOI. Repeating that release-then-patch cycle indefinitely would only ever produce archives one version behind their own citation, so this final version was instead created directly through Zenodo’s deposit API (`POST .../actions/newversion` on `22884873`), which reserves a DOI — `22885060` — *before* anything is uploaded, giving this paragraph a real, correct number to cite prior to publishing. Readers who only need a stable citation should still prefer the concept DOI above, which requires no such bookkeeping.

Sage and JSON.. The Lean displays are paired with static Sage scripts and JSON pins, not with a Sage kernel in CI (`sage_ran: false` in every certificate below): `sagemath/tate_nero_29.sage`; `sagemath/ribet_no_newforms_32.sage` with `certs/ribet_no_newforms_32.json` (`full=1, new=1, level16=0, sturm=8`); `sagemath/kolyvagin_rank0_26.sage` with `certs/kolyvagin_rank0_26.json` and `sagemath/kolyvagin_fintype_subsingleton.sage` with `certs/kolyvagin_fintype_subsingleton.json`. LMFDB labels used as pins are `32a1`, `26a1`, and `26b1` [10]. Lean verifies arithmetic involving the values these certificates carry; it does not verify their provenance, since no Sage kernel runs in CI.

Honesty note 7.2 (Certificate schema predates the `v31` rename). The two Kolyvagin JSON certificates above are immutable per-version snapshots, tagged `kolyvagin-rank0-final-v29`

and `kolyvagin-fintype-subsingleton-v28` in their own version field. Their field names `IsRankZero_is_Subsingleton`, `IsRankZero_on_26a1`, and `IsRankZero_on_26b1` predate the Lean-side rename of `IsRankZero` to `IsSubsingleton` described in Section 5; a `schema_note` field added to each certificate at v31 states this explicitly. The boolean values are unaffected by the rename, since `Subsingleton` itself did not change — only its Lean-side name did.

Repository surface. Folder READMEs are complete outside the vendor tree (difference 0 against 43735b3). GitHub About texts are 302 characters (foundations, homepage the concept DOI) and 308 characters (conjecture, homepage the version DOI). Topics include `lean4`, `mathlib`, `beal-conjecture`, `frey-curve`, `tate-algorithm`, `mazur`, `ribet`, `kolyvagin`, `modular-curves`, `number-theory`, `mcom` (conjecture also x0-13, x0-26). There is no arXiv badge.

Tags and working branches. Foundations tags v0.27–v0.31 stay live; none is deleted or force-moved to record this rename. Phase `phase-darmon-merel-4413` carries the rename as an ordinary fast-forward: `2b60ccd` (v0.30-mcom-structural-verification) → `adbca8` (Lean rename) → `b416dc9` (referee-surface script updated to check the new names) → `05fda4b` (CITATION.cff/.zenodo.json bumped to v0.31-mcom-honest-declarations, JSON certificate `schema_note` fields added), tagged v0.31-mcom-honest-declarations. Working finals from the v29 line (`tate-neron-final-v29` 11c0ad7, `mazur-no-isogeny-final-v29` 2f4e16e, `ribet-928-to-32-final-v29` 5631fd1, `kolyvagin-rank0-final-v29` fc8c7a1) are unchanged; they predate the `_final.lean` consolidation and are not themselves renamed. Conjecture `main` carries the same rename — `bac0bf7` (v30.0.0-level-26-structural-verification) → `f32a6ea` (Lean rename plus the paper’s first honesty revision) → `0367d48` (the second honesty revision, tagged v31.0.0-level-26-honest-declarations and archived by GitHub-triggered Zenodo mint as 22884682) → `b5c8911` (a DOI-backfill commit citing 22884682 and correcting the concept-DOI error, tagged v31.0.1-level-26-honest-declarations and archived as 22884873) → this revision’s own commit, which fills in the 22885060 DOI reserved for it above. Unlike its two predecessors, this commit’s tag, `v31.0.2-level-26-honest-declarations`, deliberately has *no* corresponding GitHub Release: creating one would trigger another automatic Zenodo mint of a fifth, again-one-commit-behind archive. Instead 22885060 was published directly through Zenodo’s deposit API against a source tree already containing this sentence, which is what makes it, and not 22884682 or 22884873, the correct citation for the current state of this repository. v31.0.0 and v31.0.1 stay live as historical intermediate tags and Zenodo records; nothing above is deleted or force-moved.

8. CONCLUSION

This draft records, honestly, what four Lean 4.12 files named after Tate, Mazur, Ribet, and Kolyvagin actually contain, assembled in a development aimed at the restricted gap-3 family $A^4 + B^4 = (B + 3)^{13}$: numerals and Finset cardinalities, not proofs of the classical theorems. Tate: 928 = $2^5 \cdot 29$ at the test parameters $(A, B) = (29, 1)$ only (I_0^* , I_8), not a Lean Néron conductor for general (A, B) . Mazur: 2184 and cusp card 2 versus genus 0 — no 13-isogeny is excluded. Ribet: $928/29 = 32$ with an *asserted*, not derived, matching Finset $\emptyset$ against LMFDB 32a1 of dimension 1. Kolyvagin: $|\text{Sel}_2| = 1$, $3 \cdot 7 = 21$, $L/\Omega = 1/3$ and $1/7$ on 26a1/26b1, where the repository’s surrogate `IsSubsingleton := Subsingleton` (renamed, as of this revision, from `IsRankZero`) is not the conventional algebraic-rank-zero predicate, and its application to each of these curves is false, because each curve has nontrivial torsion. Separately, and more substantively, the repository proves `no_match_32a1_of_good_red_29` (Section 4): a genuine, `native_decide`-checked mod-29 trace mismatch for gap-3 solutions with good reduction at 29, whose hypotheses are not shown to hold anywhere and which is not linked in Lean to Ribet’s level-lowering step. That lemma, not the four asserted displays, is the closest thing in this repository to a mathematically meaningful elimination result, and it is scoped to a single prime ($p = 29$), one local modulus, one newform

coefficient, and one restricted equation family (all $29^2 = 841$ residue pairs are checked, subject to filtering conditions; the scoping is in the prime and the family, not in the number of residues examined), not to Beal’s conjecture or even to the unrestricted signature $(4, 4, 13)$.

On this basis the draft is not a proof or a partial proof of any instance of Beal’s conjecture. It is a record of what a specific Lean development does and does not establish, offered so that the distinction between a displayed numeral and a mathematical theorem is not lost in translation between the repository and its description. This revision renames the misleadingly-named Lean declarations discussed above (`Tate`, `Mazur`, `Ribet`, and `Kolyvagin` displayed-data files, and `IsRankZero`→`IsSubsingleton` on `BSD_MordellWeil.lean`), with no compatibility alias kept for the removed names, and is tagged and released as a new version, `v31`, rather than silently replacing `v30`. The software versions are `v31.0.2-level-26-honest-declarations` on `beal-conjecture` (Section 7 gives the exact commit chain and its Zenodo archive, 22885060) and `v0.31-mcom-honest-declarations` on `foundations` (05fda4b, archived as 22884683). Tags `v0.27`–`v0.30` stay live and continue to refer to the pre-rename declaration names; so does the earlier `v30.0.0-level-26-structural-verification` tag on `beal-conjecture` (af520d3, About tip ea261b9, archived as 22863527), and so do the two intermediate `v31.0.0`/`v31.0.1` tags whose archives (22884682, 22884873) each cite the DOI of the record *before* them rather than their own, for the reason given in Section 7. Parent algebraic statements remain `def Prop`. No remaining timers.

The companion foundations end-of-proof and post-final releases are `v30.1.7-beal-final-43735b3` (10.5281/zenodo.22912430) and `v25.0.1-forward-43735b3` (10.5281/zenodo.22922473). They preserve the forward numerical surface while distinguishing the proved $B < 1000$ elimination from the explicit full $B \leq 10^6$ search and the separate general Beal common-prime boundary.

This draft is a Mathematics of Computation submission, not an arXiv preprint. Cite this revision’s version DOI 10.5281/zenodo.22885060 and the `beal-conjecture` concept DOI 10.5281/zenodo.22041831.

REFERENCES

1. John E. Cremona, *Algorithms for modular elliptic curves*, 2 ed., Cambridge University Press, 1997.
2. David Fox, *beal-conjecture (concept)*, Zenodo, 2025, Concept DOI; `isVersionOf` for every `beal-conjecture` version record, including `v30` (10.5281/zenodo.22863527) and `v31` (10.5281/zenodo.22884682, 10.5281/zenodo.22884873, 10.5281/zenodo.22885060). Distinct from the `beal-level-26-foundations` concept DOI 10.5281/zenodo.22379293, which archives that separate repository’s own releases.
3. ———, *beal-level-26-foundations (concept)*, Zenodo, 2025, Concept DOI for the separate `beal-level-26-foundations` repository; `isVersionOf` its `v0.31` record 10.5281/zenodo.22884683.
4. ———, *beal-conjecture: Formal verification of the level 26 foundations for signature (4,4,13)*, Zenodo, 2026, Version `v31.0.2-level-26-honest-declarations`; new version of 10.5281/zenodo.22884873; archives the re-named declarations described in this revision. Unlike 10.5281/zenodo.22884682 and 10.5281/zenodo.22884873, this DOI was reserved through Zenodo’s deposit API and published only once the source tree (including this citation) matched the reservation, so this record correctly cites itself.
5. ———, *beal-conjecture: Formal verification of the level 26 foundations for signature (4,4,13)*, Zenodo, 2026, Version `v30.0.0-level-26-structural-verification`; new version of 10.5281/zenodo.22854603 and 10.5281/zenodo.22851334; archives the pre-rename declaration names.
6. ———, *beal-level-26-foundations: final beal forward interface*, Zenodo, 2026, Version `v30.1.7-beal-final-43735b3`; full search and general common-prime theorem retained as explicit proof boundaries.
7. ———, *beal-level-26-foundations: post-final v25 forward retrofit*, Zenodo, 2026, Version `v25.0.1-forward-43735b3`; Matveev and level-26 forward order with the full $B \leq 10^6$ search retained as an explicit boundary.
8. Gerhard Frey, *Links between stable elliptic curves and certain Diophantine equations*, Annales Universitatis Saraviensis. Series Mathematicae **1** (1986), no. 1, 1–40.
9. Victor A. Kolyvagin, *Euler systems*, The Grothendieck Festschrift, Vol. II **87** (1990), 435–483.
10. LMFDB Collaboration, *The L-functions and modular forms database*, <https://www.lmfdb.org>, 2026, Labels 32a1, 26a1 (LMFDB 26.a2), 26b1 (LMFDB 26.b2).

11. R. Daniel Mauldin, *A generalization of Fermat's last theorem: The Beal conjecture and prize problem*, Notices of the American Mathematical Society **44** (1997), no. 11, 1436–1437.
12. Barry Mazur, *Rational isogenies of prime degree*, Inventiones Mathematicae **44** (1978), no. 2, 129–162.
13. Kenneth A. Ribet, *On modular representations of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ arising from modular forms*, Inventiones Mathematicae **100** (1990), no. 2, 431–476.
14. Jean-Pierre Serre, *Sur les représentations modulaires de degré 2 de $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$* , Duke Mathematical Journal **54** (1987), no. 1, 179–230.
15. John Tate, *Algorithm for determining the type of a singular fibre in an elliptic pencil*, Modular Functions of One Variable IV, Lecture Notes in Mathematics, vol. 476, Springer, Berlin, 1975, pp. 33–52.

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